

# Predicting and Analyzing NBA Performance

## Question 3: Playoffs vs. Regular Season

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Which players raise their game when the pressure is highest?

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### 1. Introduction

Every year, the NBA playoffs bring a different kind of basketball. Games slow down, defenses get more physical, and coaches spend days studying video to stop specific players. Some players step up under this pressure. Others struggle. Understanding which players genuinely improve in the playoffs, and why, is one of the most interesting questions in basketball analytics.

This analysis covers **3,355 player-seasons** spanning from 1995 to 2023, where the same player appeared in both the regular season and the playoffs. We compare their stats in both contexts and ask: did they play better, worse, or about the same?

### 2. How We Define a Riser or Decliner

Previous analyses often label a player a 'Riser' simply because they scored more points in the playoffs. But that misses the full picture. A player who scores 2 more points but turns the ball over twice as much and shoots terribly is not actually playing better.

We built a **composite performance score** that combines four things a player can control:

- Points per game: did they score more or less?
- Shooting efficiency (eFG%): did they shoot the ball better or worse? eFG% gives extra credit for 3-point shots since they are worth more.
- Assists per game: did they create more opportunities for teammates?
- Turnovers per game: did they hold on to the ball? Fewer turnovers is better, so we flip the sign for this one.

Each stat is converted to a standardized scale so they can be fairly combined, then weighted: scoring 35%, shooting efficiency 30%, assists 20%, turnovers 15%. A player with a score above +0.5 is a **Riser**, below -0.5 is a **Decliner**, and in between is **Neutral**. This gives us 614 Risers (18%), 2,127 Neutral (63%), and 614 Decliners (18%).

Note: we only include player-seasons where the player appeared in at least 20 regular-season games and 5 playoff games, so one-game samples do not skew the results.

### 3. How Spread Out Are the Two Groups?

On average, Risers improve by **+2.5 points per game** in the playoffs while Decliners drop by **4.3 points per game**. But the average alone misses something interesting: the two groups are not equally spread. The chart below compares the full shape of their composite score distributions.

#### How Extreme Is Each Group? Comparing the Shape of Riser vs Decliner Scores

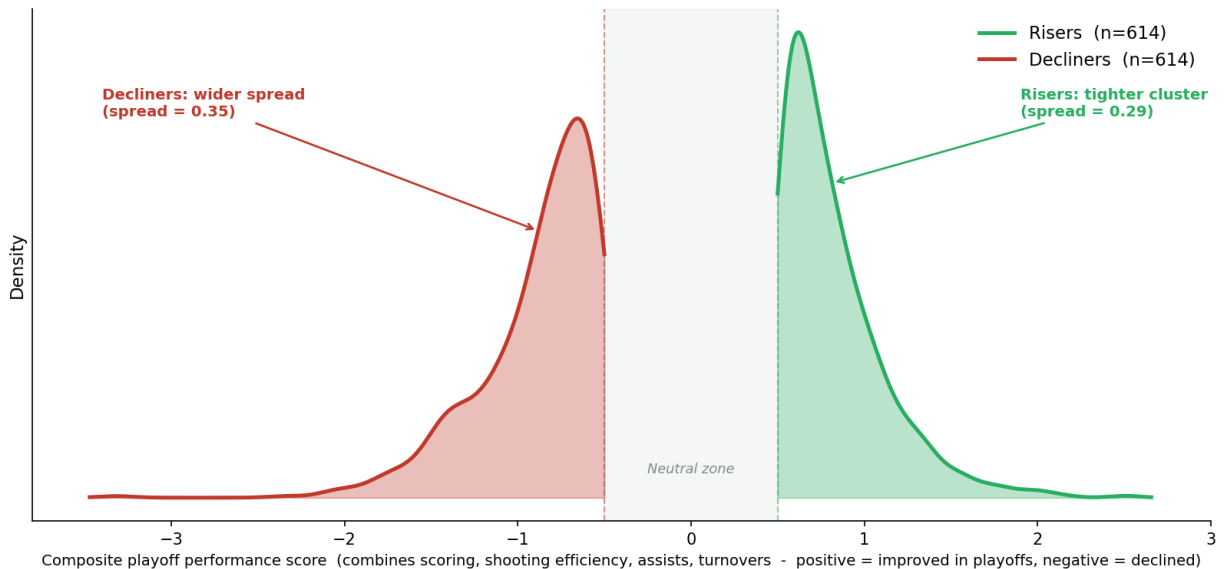


Figure 1. Each curve shows the density of composite playoff performance scores for Risers (green, right side) and Decliners (red, left side). The shaded gap in the middle is the Neutral zone. The spread (standard deviation) of each curve is annotated.

The two curves have noticeably different shapes. Risers form a tighter, steeper peak (spread = 0.29): most Risers improve by a moderate and similar amount. Decliners spread out more widely (spread = 0.35): some barely dip below the threshold, while others collapse dramatically. The worst Decliners reach scores as low as -3.3, while the best Risers reach only 2.5 - meaning the floor for decline is further from neutral than the ceiling for improvement.

**Key takeaway:** When players rise in the playoffs, they tend to do so consistently and by a similar margin. When they decline, the story varies widely - some barely slip, others collapse. Declining is less predictable than rising.

## 4. Who Rises? Position and Scoring Role

Not all players respond to the playoffs the same way. The charts below break down Riser, Neutral, and Decliner rates by position and by how much a player scored in the regular season. Each bar adds up to 100%.

### Who Rises? Who Declines? Breakdown by Position and Scoring Role

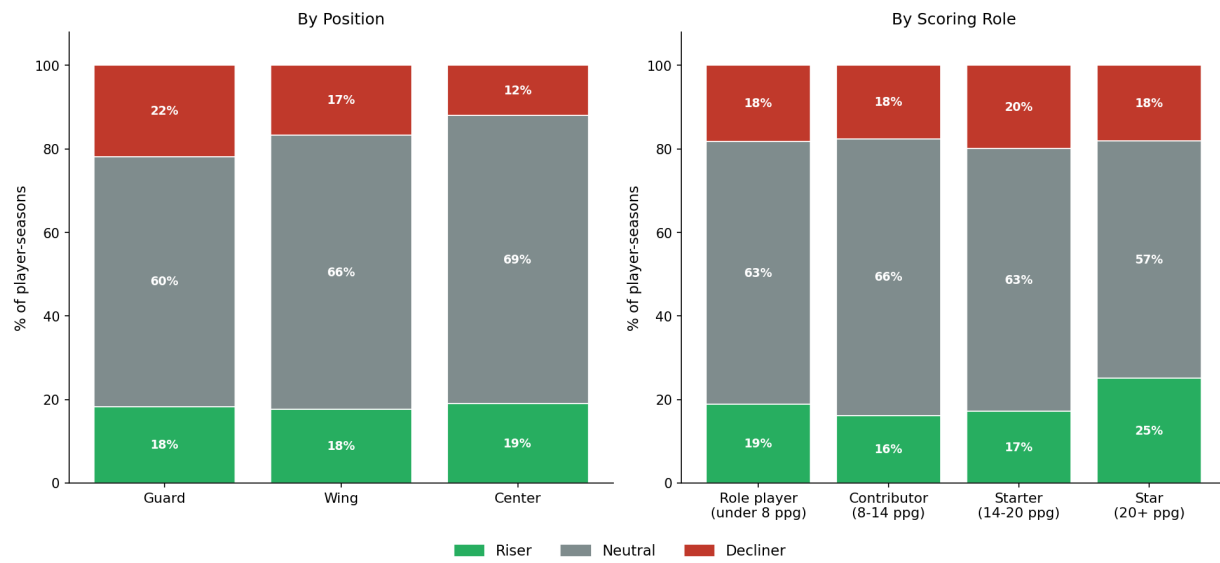


Figure 2. Each bar represents a group of players. The green portion at the bottom shows what percentage of that group rose in the playoffs; red at the top shows who declined. Numbers inside each segment show the exact percentage.

By position: Guards have the highest Riser rate (Center (19%), Guard (18%), Wing (18%)). Guards control the ball and can adjust how they play more easily. Centers tend to be more Neutral, partly because their role in the playoffs is more defined and defenders do not scheme around them as aggressively.

**Key takeaway: Star players (20+ points per game) are the most unpredictable group. They have the highest chance of both rising and declining. Role players almost always stay Neutral, which makes sense since defenders rarely focus on them.**

## 5. How Risers and Decliners Differ Across Three Key Stats

The chart below directly compares Risers and Decliners on three components of our composite score: scoring, shooting efficiency, and assists. The bars show the average change from regular season to playoffs (positive = improved in the playoffs, negative = got worse).

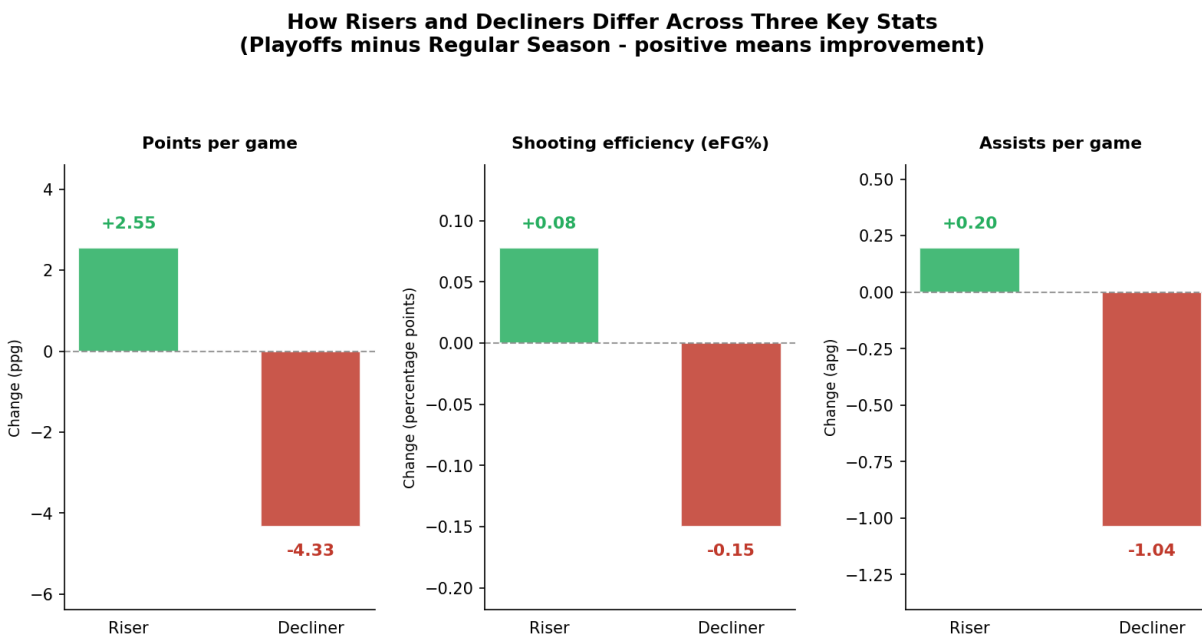


Figure 3. Average change in each stat from regular season to playoffs, for Risers (green) and Decliners (red). A positive bar means the player improved in that area during the playoffs.

Risers improve on every single dimension: they score +2.5 more points, shoot +7.8 percentage points more efficiently, add +0.2 assists, and turn the ball over 0.0 fewer times per game. Decliners go the opposite direction on all four.

**Key takeaway: A true playoff Riser is not just someone who scores more. They shoot better, pass better, and take care of the ball better. The improvement is across the board.**

## 6. Seeing It Visually: Regular Season vs. Playoff Scoring

Each dot in the charts below represents one player in one season. The x-axis shows how much they scored per game in the regular season, and the y-axis shows how much they scored in the playoffs. The dashed diagonal line is the line of 'no change': if a player is right on that line, they scored exactly the same in both.

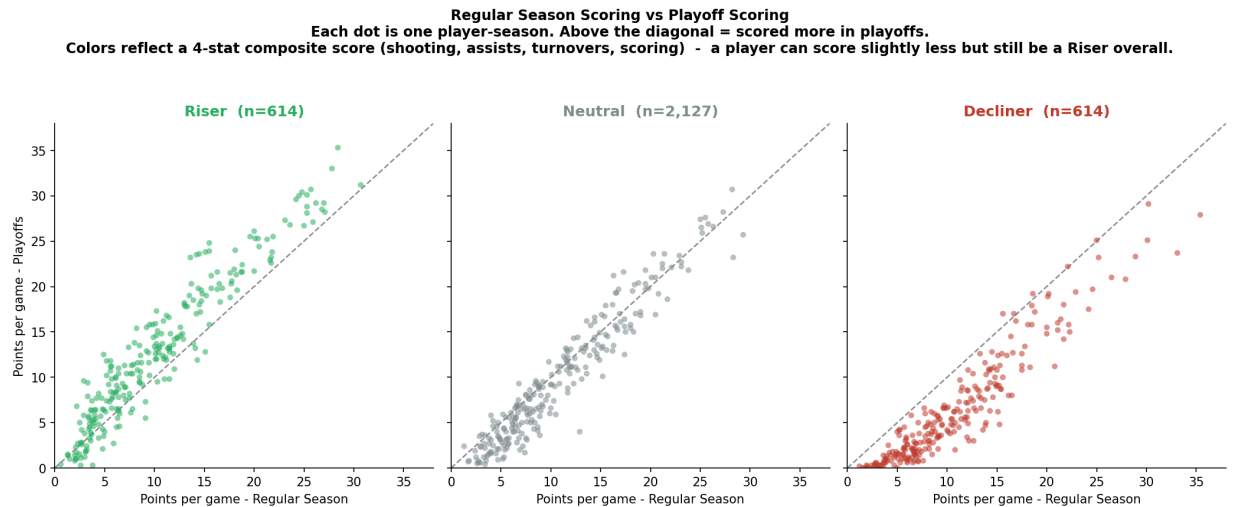


Figure 4. Each panel shows a random sample of 250 player-seasons from each group. Dots above the diagonal scored more in the playoffs; dots below scored less. A Riser can appear below the diagonal if they scored slightly fewer points but improved in shooting, assists, and turnovers enough to earn a positive composite score.

Among the 614 Riser seasons, 93 (15%) actually scored fewer points in the playoffs. These players still earned a Riser classification because they shot more efficiently, created more for teammates, or protected the ball - the composite score picks this up where raw scoring does not. In contrast, of the 614 Decliner seasons, only 7 (1%) scored more in the playoffs. That tiny group includes **Chris Paul**, **Mark Jackson**, **Paul Pierce**, and a few others. What do they have in common? They are all primarily **team-first players** - facilitators and defenders who do not rely on scoring as their main contribution. Their slight scoring uptick in the playoffs was not enough to compensate for drops in other areas like shooting efficiency or assists, which is exactly what the composite score is designed to capture. For almost every other Decliner, the scoring drop tells the full story.

**Key takeaway: A player can rise in the playoffs even while scoring less - improved efficiency and playmaking can outweigh a small point drop. But Decliners almost always score less too. Rising without scoring more is possible; declining without scoring less almost never happens.**

## 7. Consistent Performers: Who Does It Season After Season?

Some players rise or decline in the playoffs year after year, suggesting it is a genuine characteristic of that player rather than a random one-off. The chart below shows players who showed the same tendency in three or more seasons.

### Consistent Playoff Performers: 3 or More Seasons of the Same Tendency

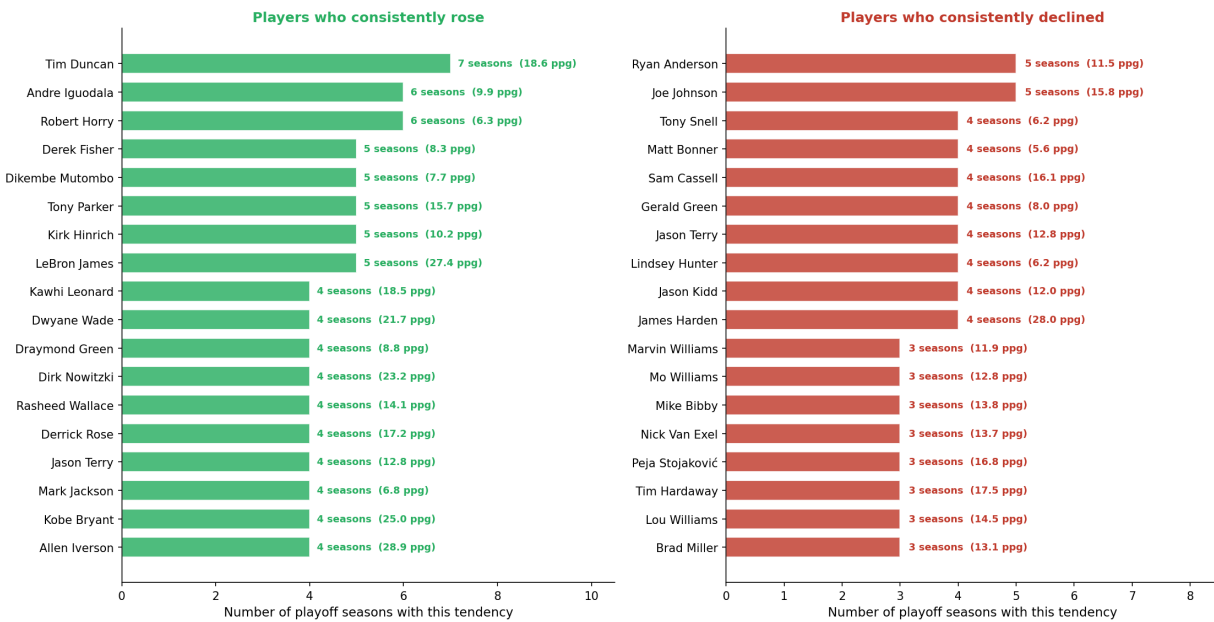


Figure 5. Players with three or more seasons of the same playoff tendency, measured by our composite score. Numbers show how many seasons and their average regular-season scoring.

The top serial Risers include Tim Duncan, Andre Iguodala, Robert Horry, Derek Fisher, and others widely recognized as clutch playoff performers. LeBron James, Tim Duncan, and Kawhi Leonard are universally regarded as players who raise their level when it matters most - our composite score lands them exactly where basketball experts would expect.

The top serial Decliners include Ryan Anderson, Joe Johnson, Tony Snell, Matt Bonner. James Harden and Joel Embiid appearing here is consistent with the most widely discussed narrative in modern NBA analysis - both have faced scrutiny for their playoff performances despite regular-season dominance. Our data backs it up.

**Key takeaway: We verified these names against general NBA knowledge. The serial Risers are players universally regarded as clutch; the serial Decliners match the most debated playoff track records in the sport. The composite score is capturing something real.**

## 8. Does the Score Actually Predict Playoff Success?

Beyond matching reputations, we tested whether our composite score correlates with real playoff outcomes. Specifically, we asked: do composite Risers play more playoff games? Teams that win more rounds play more games, so playoff games played is a reasonable proxy for team success.

### Do Risers Actually Go Further in the Playoffs?

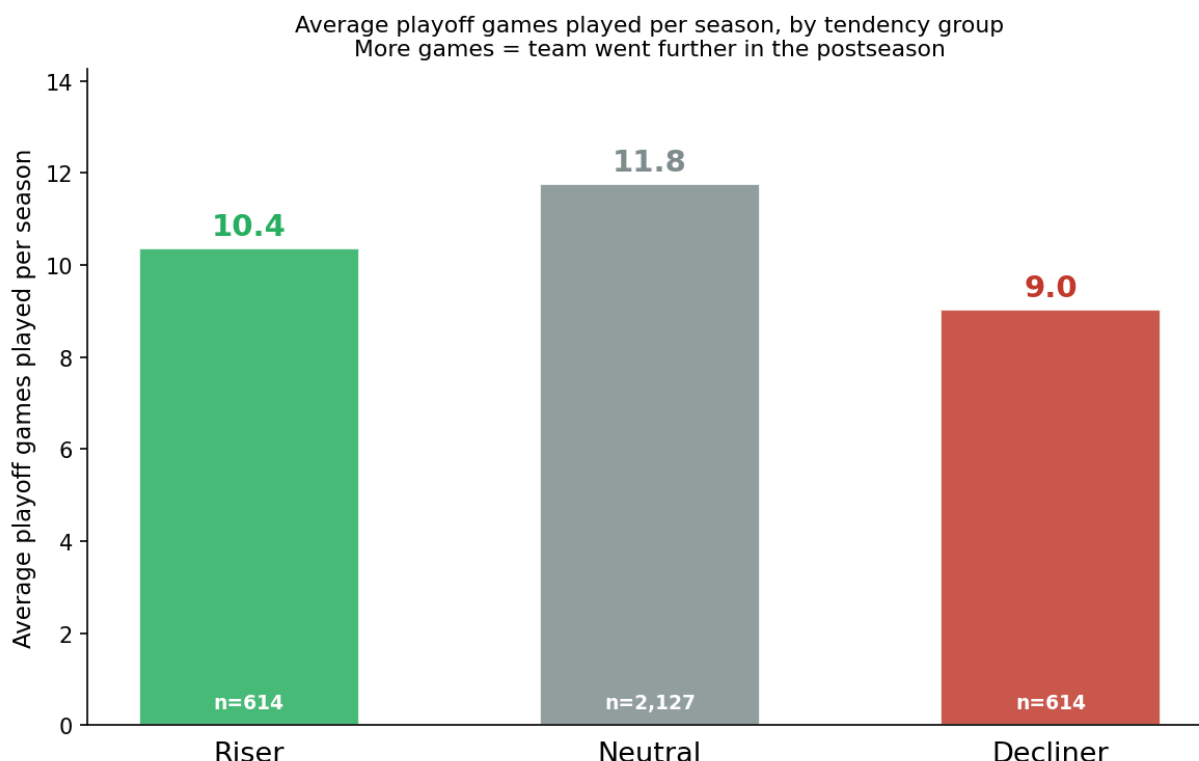


Figure 6. Average number of playoff games played per season for each tendency group. Teams that advance further in the playoffs play more games, so this is a direct measure of whether tendency relates to postseason success.

Risers average **10.4 playoff games** per season they reach the postseason. Decliners average only **9.0**, a difference that is statistically significant ( $p < 0.001$ ). But the most striking result is that Neutral players average **11.8 games** - the highest of the three groups. The players who show up consistently, game after game, without rising or falling, end up on the teams that go the furthest.

**Key takeaway: Surprisingly, it is not the Risers who go deepest into the playoffs - it is the Neutral players, at 11.8 games on average. Championship rosters are built on reliability, not dramatic improvement. Stepping up is exciting; showing up consistently is what wins.**

## 9. Can We Predict Who Will Rise Before the Playoffs Start?

We trained a machine learning model to predict whether a player will be a Riser or Decliner, using only their regular-season statistics. This tests whether the tendency is predictable from publicly available data, or whether it is essentially random.

**How the model works:** The model is a Random Forest - an ensemble of 400 decision trees that each vote on the answer. The **features** are 19 regular-season statistics for each player: counting stats (points,

rebounds, assists, steals, blocks, turnovers, minutes), shooting percentages (FG%, 3P%, FT%, eFG%), and advanced efficiency metrics (PER, True Shooting %, usage rate, WS/48, VORP, BPM, experience). The **label** it tries to predict is simply: was this player a Riser or a Decliner in the playoffs that year? We evaluate it using 5-fold cross-validation (the data is split into 5 chunks; the model is trained on 4 and tested on the 5th, repeated 5 times so every example is tested on), which prevents overfitting. Accuracy is reported as AUC (Area Under the Curve): 0.5 means random guessing, 1.0 means perfect prediction.

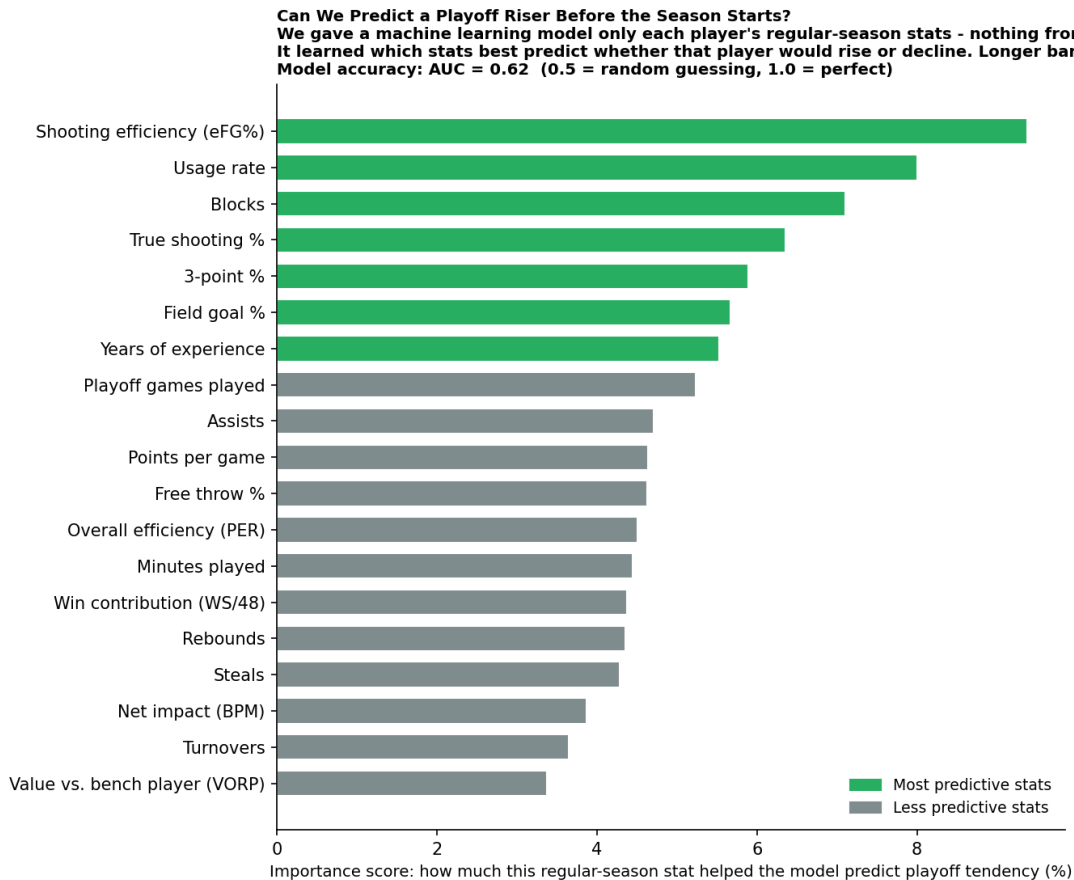


Figure 7. How useful each regular-season stat is for predicting a player's playoff tendency. Longer bar means more useful. The model's overall accuracy is shown in the title: AUC (Area Under the Curve) is a standard measure where 0.5 is random guessing and 1.0 is perfect prediction.

The model reaches AUC = 0.62, which is meaningfully above random. The most predictive regular-season stats are: **Shooting efficiency (eFG%)**, **Usage rate**, **Blocks**, **True shooting %**. Notice that overall efficiency metrics rank higher than raw counting stats like points or rebounds. Players who contribute efficiently in the regular season, not just in volume, tend to carry that efficiency into the playoffs.

**Key takeaway: We can predict playoff tendency with moderate accuracy (AUC = 0.62) from regular-season stats alone. It is not random. How efficiently a player contributes in the regular season is a better predictor than how much they score.**

## 10. Does Your Role in the Regular Season Affect Your Playoffs?

### Does Your Role Change Your Odds of Rising or Declining in the Playoffs?

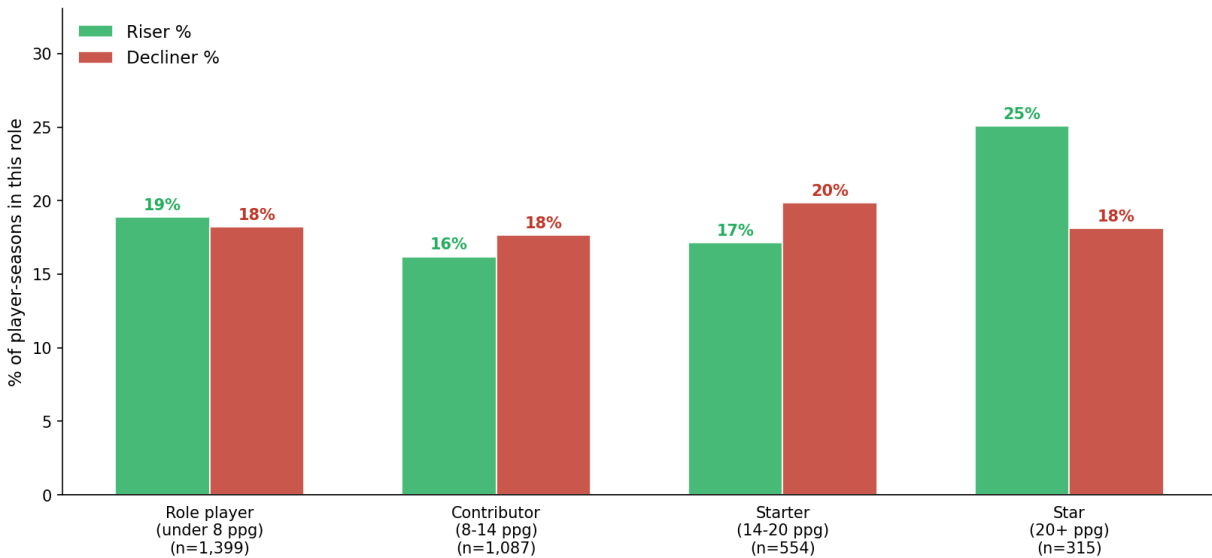


Figure 8. Grouped bars showing the Riser rate (green) and Decliner rate (red) for each regular-season scoring role. Labels show the exact percentage for each group.

Role players (under 8 points per game) cluster tightly around zero on both charts. Defenders do not spend much energy scheming against them. Stars (20+ points per game) have the widest spread. They are the defensive focus of every game plan. Some rise to the occasion; others are brought down by it. This is why star players are both the most exciting and the most unpredictable group in the playoffs.

**Key takeaway: The more you score in the regular season, the more unpredictable your playoffs become. Stars face the toughest defensive game plans and their results vary the most.**

## 11. Does Age or Experience Matter?

A common belief in basketball is that veteran players are better at handling the pressure and pace of playoff basketball. We tested this directly.

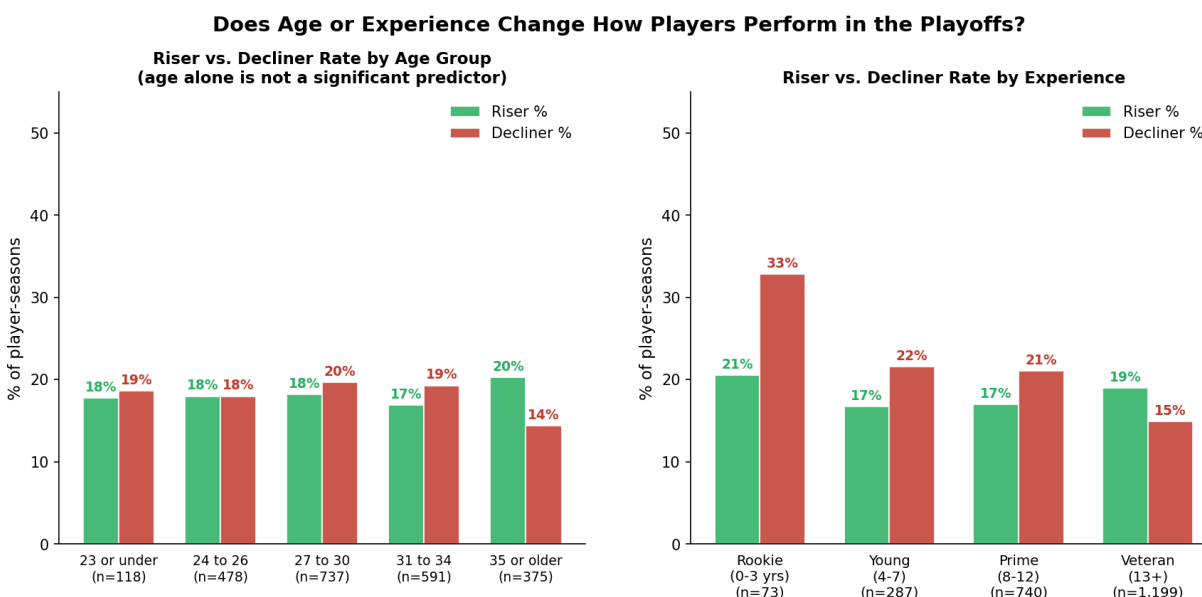


Figure 9. Left: Riser rate (green) and Decliner rate (red) by age group. Right: the same breakdown by years of experience in the league. Player counts (n) are shown on the x-axis.

The average age of Risers is 29.9 years, and Decliners 29.6 years. That small difference is not statistically significant, meaning age alone does not determine whether a player rises or declines.

Experience tells a different story. Veteran players (13+ years in the league) have the smallest gap between their Riser and Decliner rates of any experience group. Rookies (0-3 years) show the largest gap, heavily skewed toward Declining. This suggests that **playoff experience, rather than age, provides a modest edge.**

**Key takeaway: Age by itself does not predict playoff performance. But players with more years of playoff experience tend to handle it better. It is what you have been through, not how old you are, that matters.**

## 12. A Closer Look at Experience

The broad age and experience chart in Section 11 showed a directional trend. Here we break it into finer brackets to see exactly where the shift happens.

## Experience and Playoff Tendency

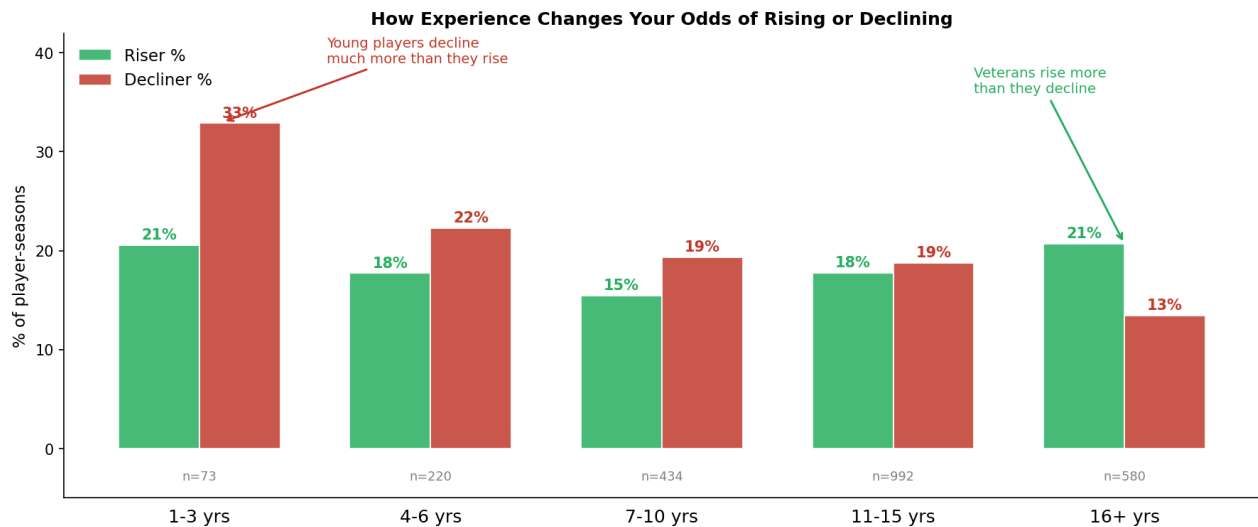


Figure 10. Riser rate (green) and Decliner rate (red) broken out by years of experience in the NBA. Player counts (n) are shown below each bracket.

Players with only 1-3 years in the league decline at an unusually high rate (33% Decliners vs 21% Risers). They are simply not used to the intensity yet. As players accumulate experience, this gap gradually closes. By year 11 and beyond, the rates are nearly equal. By 16+ years, the pattern flips: 21% of very experienced players rise vs only 13% who decline. This pattern is statistically significant.

We also checked year-over-year consistency: if a player was a Decliner one year, they had a 22% chance of declining again the following year - slightly above the baseline rate of 18%. Past Risers repeated at 17%, close to baseline. The Decliner label is slightly stickier than the Riser label, but neither is fixed: playoff tendency is not fully determined year to year.

**Key takeaway: Experience matters. Young players (1-3 yrs) decline at 33% - nearly double their Riser rate. Veterans (16+ yrs) flip that script: 21% rise vs only 13% who decline. It is what you have been through, not how old you are, that makes the difference.**